COMP20008 Elements of Data Processing

Lecture 4: Data scaling, imputation, and sampling

(“the missing data and the importance of representative samples”)

Semester 2 2026
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Announcements

CALL FOR PARTICIPATION

http://www.alta.asn.au/events/sharedtask2026

The Australasian Language Technology Association (ALTA) is organising a programming competition for university undergraduate and postgraduate students.

All participants compete to solve the same problem. The problem highlights an active area of research and programming in the area of language technology.

This year's shared task is: **classify the sentiment and the sarcasm of Australian and British English text**

The tentative key dates are:

Right Now - Registration and release of training and development data
22 Sep 2026 - Release of test data
28 Sep 2026 - Deadline of submission of runs
01 Oct 2026 - Notification of results
26 Oct 2025 - Deadline of submission of system description
30 Nov - 2 Dec 2026 - Presentation of results at ALTA 2026

Assessing Data Quality

Your input

	Date of Birth
1	20 years ago
2	-
3	20/5/79
4	13th Feb. 2019
5	-
6	11/20/66
7	1961年8月4日

Things you must check

Accuracy
- Correct or wrong, accurate or not

Completeness
- Not recorded, unavailable

Consistency
- E.g. discrepancies in representation

Timeliness
- Updated in a timely way

Believability
- Do I trust the data is true?

Interpretability
- How easily can I understand the data?

Lecture outline

- **Scaling**
- Imputation
- Sampling

Resolving Inconsistencies

The Chaos (Raw Input)		The Standard (Refined Output)
Naming	"three" "3" 3	3
Formats	"3/4/2016" "3rd April 2016" "4/3/16" "2016-04-03"	"2016-04-03"
Equivalency	Age=20 Birthdate="1/1/1971"	Unified Age Variable
Duplication	**Two students** with the same student id	Data Cleaning (Removing Errors)
Outliers	62,72,75,75,80,82,84, 86,87,87,89,89,90,**9999**	???????

Resolving Inconsistencies

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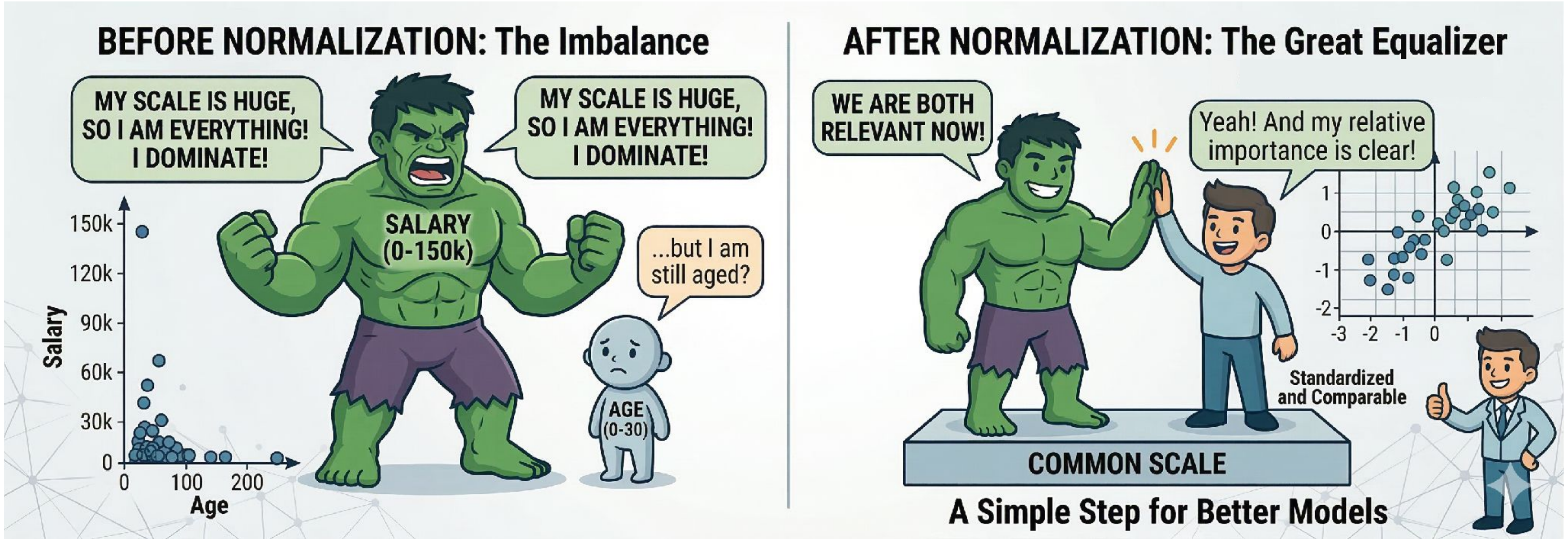
Duplication	**Two students** with the same student id	Data Cleaning (Removing Errors)
Outliers	62,72,75,75,80,82,84, 86,87,87,89,89,90,**9999**	Is 9999 an error? **Domain knowledge is required.** If this is a list of <u>hospital patient ages</u>, 9999 is corrupt data. If it is a list of <u>Instagram connections</u>, 9999 is perfectly valid.

Feature scaling

You need different sets of numbers to be in the same range so you can compare them

Important pre-processing step in the pipeline



Rescaling Example

Who flies further for its size — an eagle or a wren?

The Imbalance

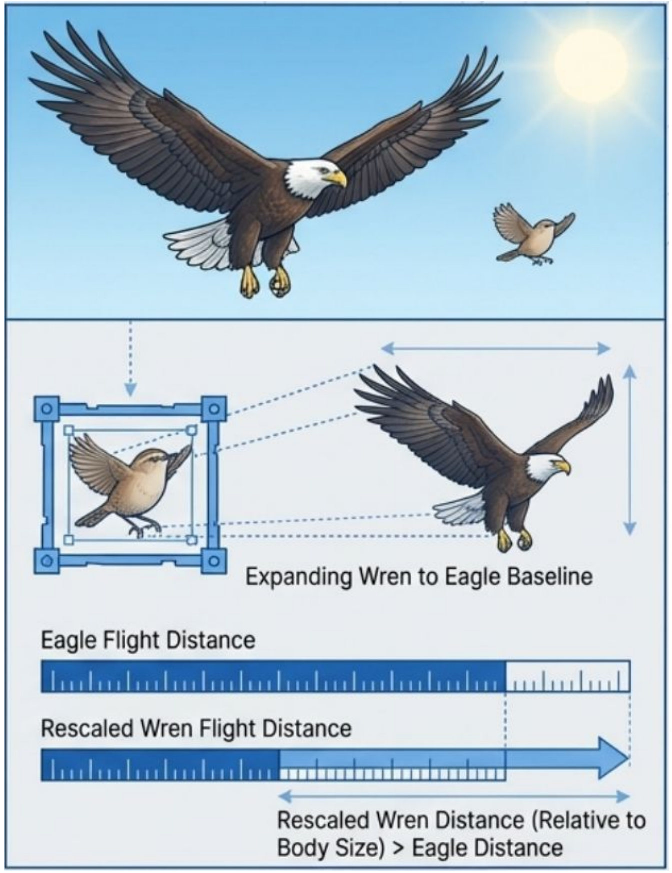
Cannot directly compare their absolute flight distances because their baseline capabilities (body size) are vastly different.

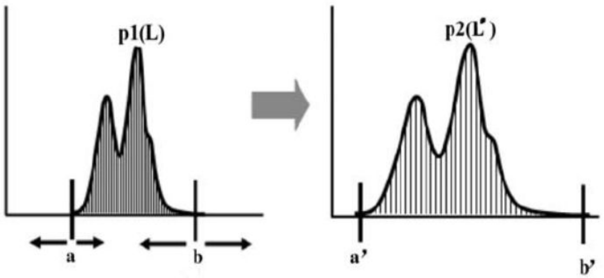
The Transformation

Must mathematically shrink the eagle or expand the wren so they share the exact same scale of body size.

The Comparison

Once the baselines are aligned, the distance metric becomes a true reflection of relative performance.

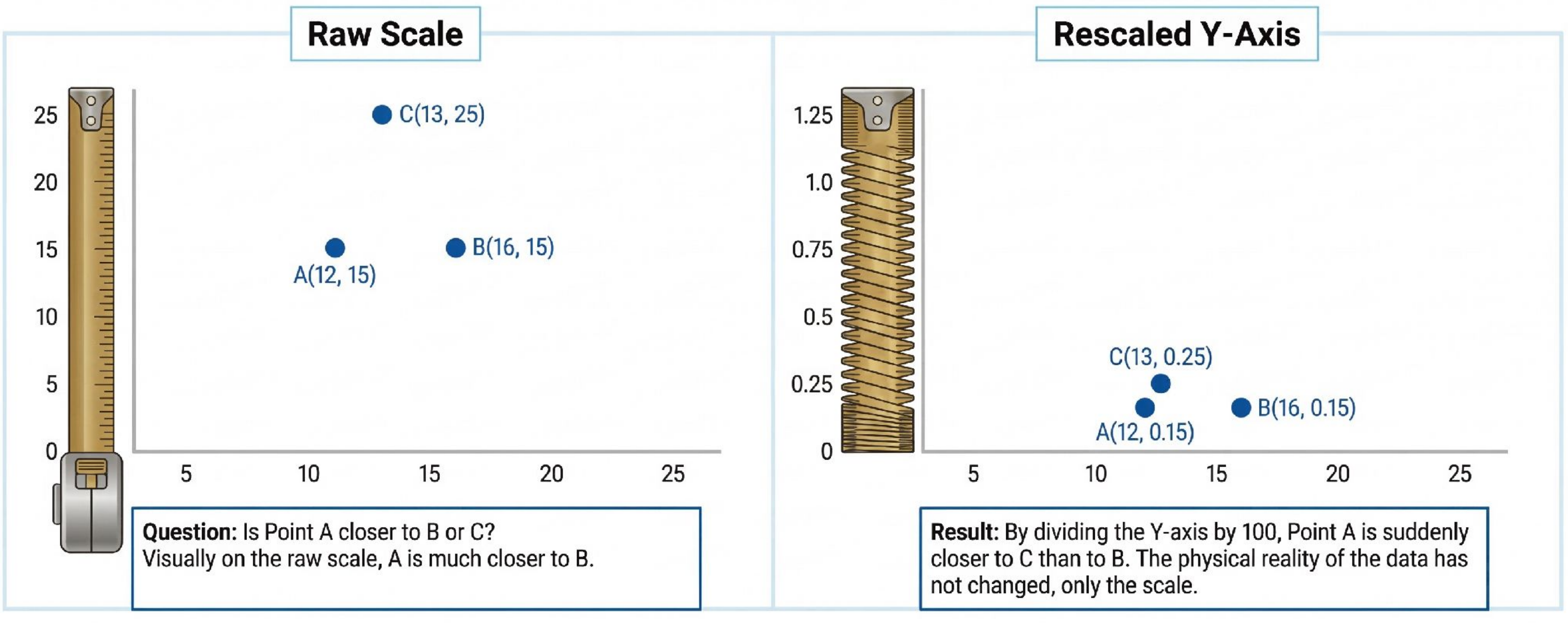




https://www.researchgate.net/publication/236853

The Mathematical Imperative of Rescaling

Algorithms rely on distance metrics to find relationships.
If variables operate on completely different scales, the variable with the larger absolute numbers will artificially dominate the algorithm.



Feature scaling

It is essential for many machine learning models

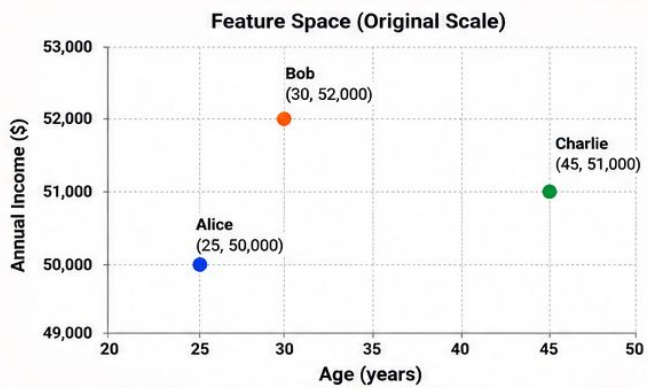
1 WITHOUT SCALING

Example Dataset

Customer	Age (years)	Annual Income ($)
Alice	25	50,000
Bob	30	52,000
Charlie	45	51,000

Age range: 25 – 45 (range = 20)

Income range: 50,000 – 52,000 (range = 2,000)



Distance between Alice and Bob

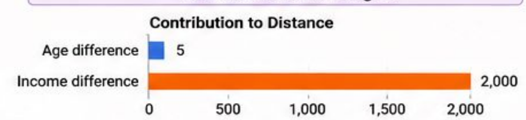
$$d = \sqrt{(30-25)^2 + (52,000-50,000)^2}$$
$$= \sqrt{5^2 + 2,000^2}$$

Contribution to squared distance:

Age part: $(5)^2 = 25$ (0.0006%)

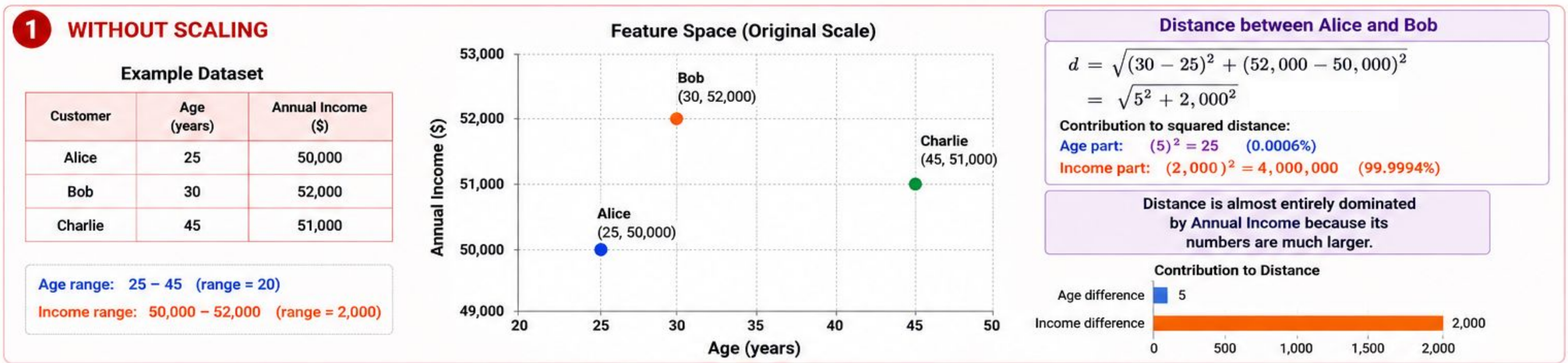
Income part: $(2,000)^2 = 4,000,000$ (99.9994%)

Distance is almost entirely dominated by Annual Income because its numbers are much larger.



Feature scaling

It is essential for many machine learning models



Critical for both distance-based ML approaches such as k-NN, k-means as well feature-based ones such as SVMs (including NNs trained with gradient descent)

Two rescaling methods

Min–max normalisation rescales values to a **fixed interval**, usually from **0 to 1**:

$$(x-x_{min})/(x_{max}-x_{min})$$

Use when numbers have different units (*inches*, *cm*, *etc.*)

Standardisation (z-score) centers the data to have a mean of 0 and a standard deviation of 1

$$z = (x-\mu)/\sigma$$

μ, σ – *mean and standard deviation*

No fixed range; Useful to compare distributions

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$$z = (x - \mu) / \sigma$$

How far is the value from the mean?

μ, σ – *mean and standard deviation*

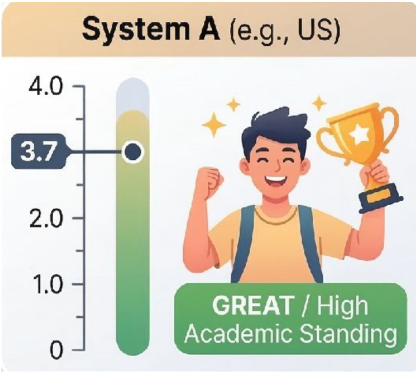
No fixed range; Useful to compare distributions

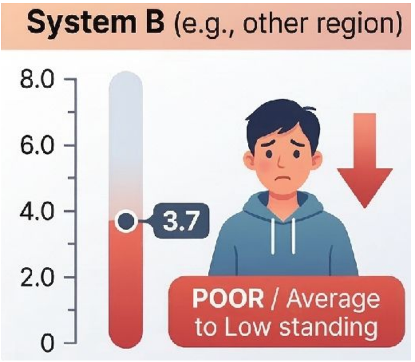
Case Examples: Normalisation

Case 1: Student scores: GPA (Requires Normalisation)

The problem:
University A grades out of **4.0**.
University B grades out of **8.0**.
A score of 3.7 means entirely different things.

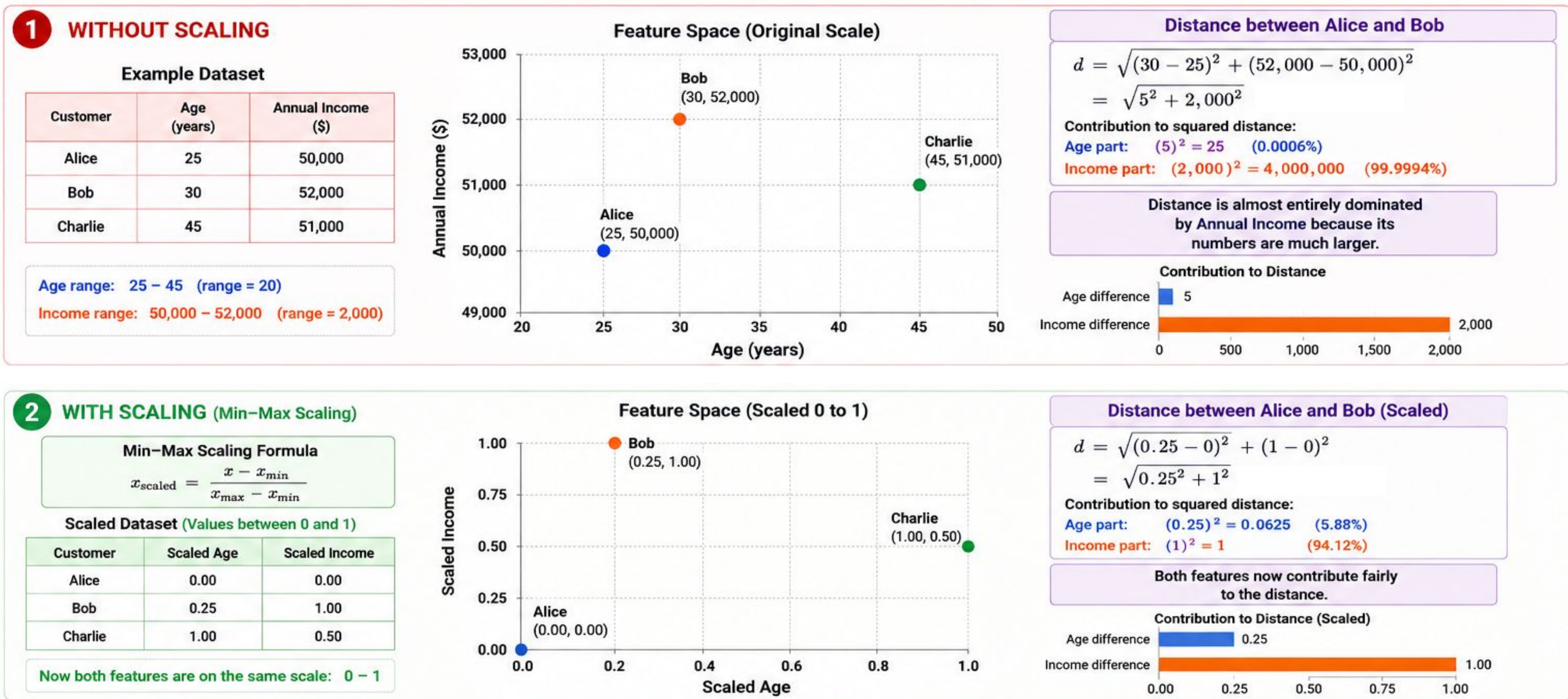
The solution:
To directly compare the individual students, we normalise the scores to a percentage or fraction (**3.7/4**=0.925 vs **3.7/8**=0.463).





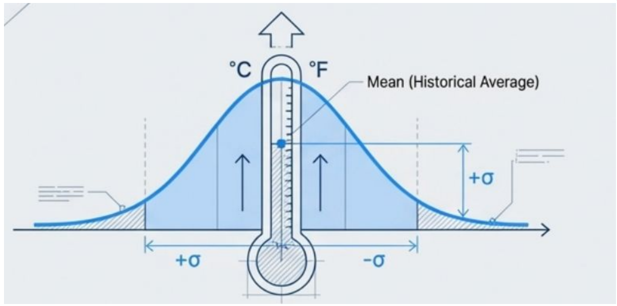
Feature scaling

It is essential for many machine learning models



Case Examples: Standardisation

Case 2: Climate Temperatures
(Requires Standardisation)



The problem:
Comparing Celsius [0–100] vs Fahrenheit [32–212].
While scales differ, weather has no fixed maximum endpoint to safely divide by.

The solution:
To determine which country had a relatively more severe summer compared to its own historical average, we standardise the **temperature values**.

This converts temperatures into **how far they deviate from the historical average**, allowing us to compare relative heat intensity across countries, rather than comparing the raw degrees.
z = (temperature − average) / standard deviation

Normalisation vs Standardisation

Normalisation

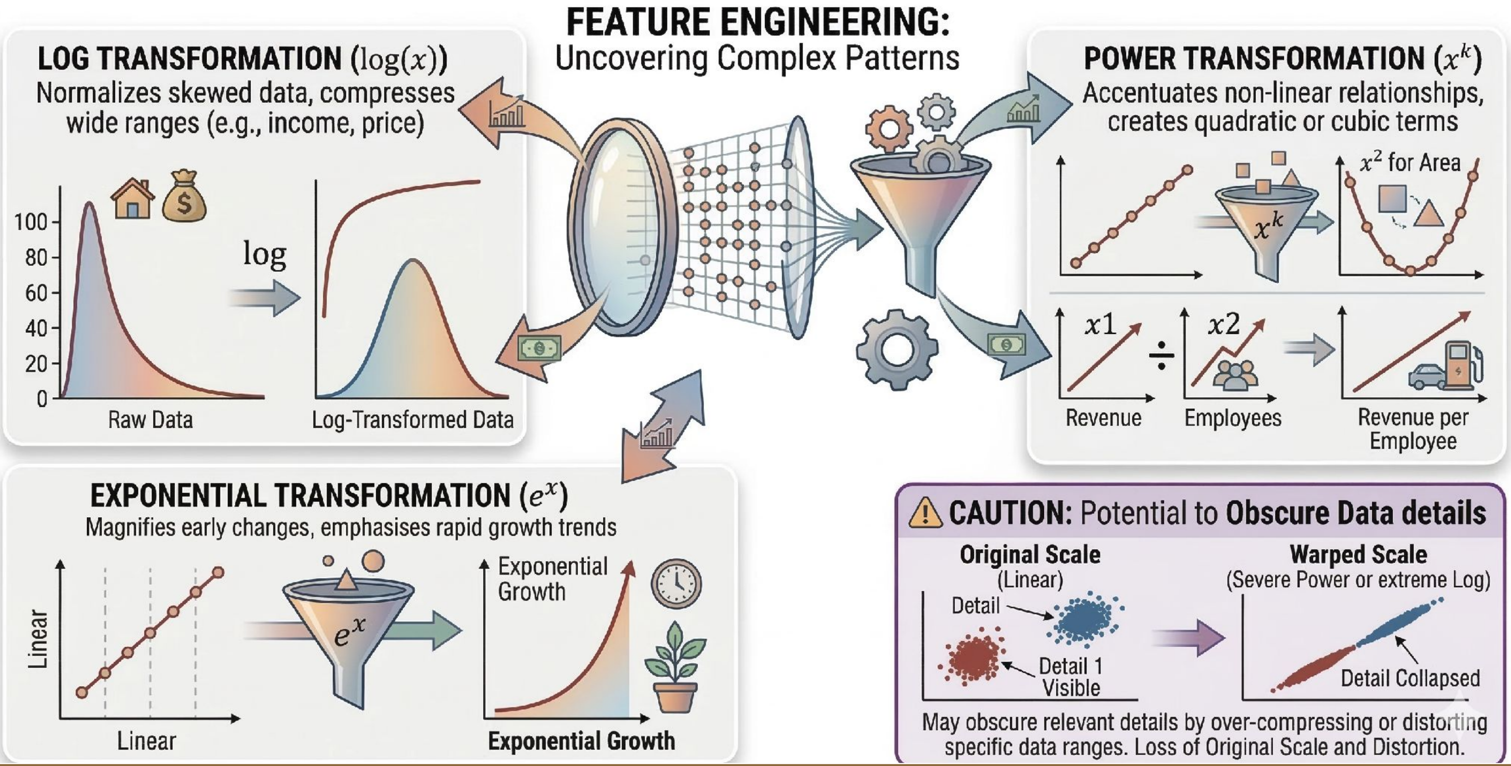
- Helpful for comparing individual values
- Scales the models using minimum and maximum values
- Used when data are on various scales
- Values reset to fall between fixed end points [0, 1] or [-1, 1]

Standardisation

- Helpful for comparing value distributions
- Scales the models using the mean and standard deviation
- Useful when data are open-ended
- Value ranges' mean is set to 0 and its standard deviation to 1

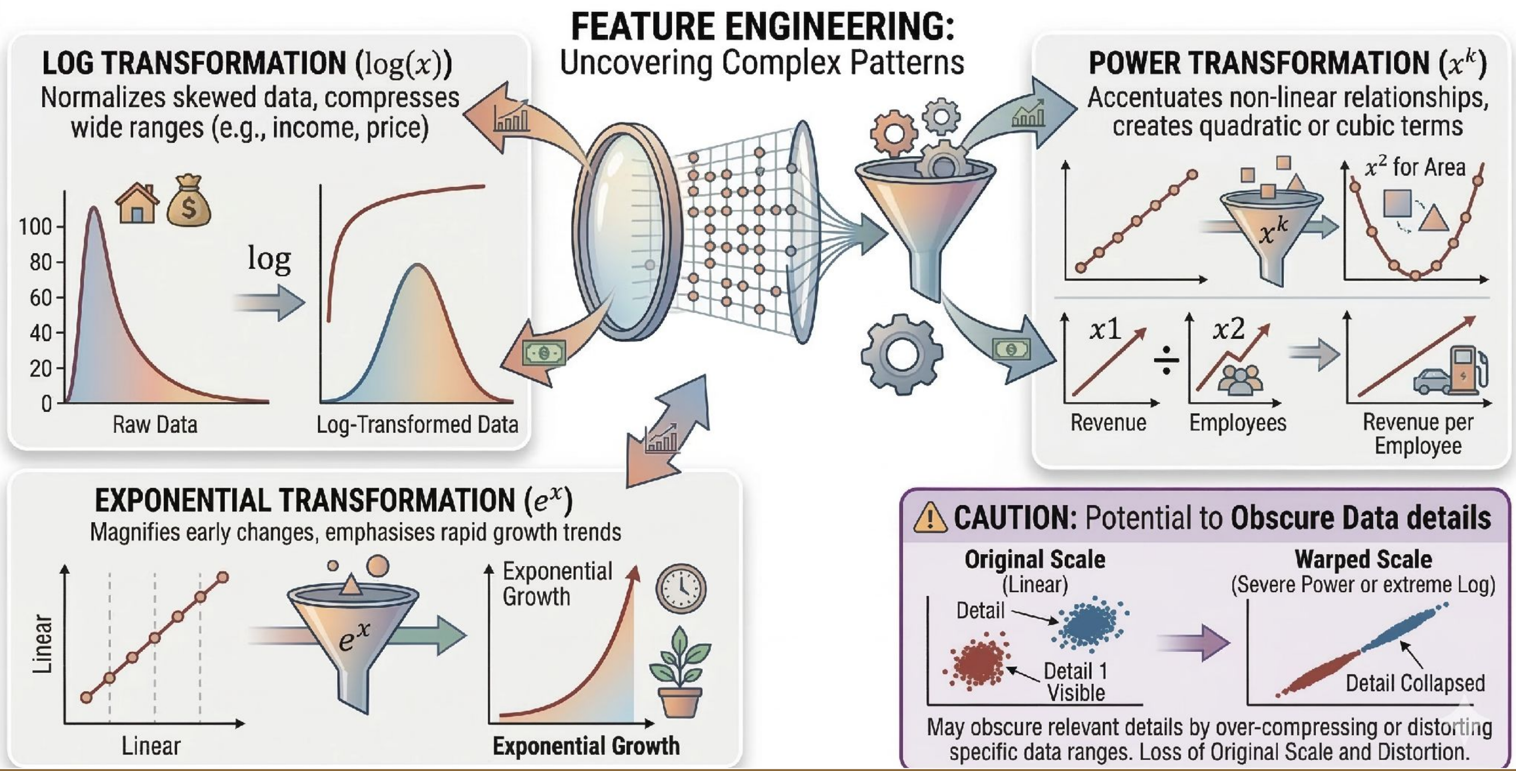
Other rescaling transformations

More scaling transformations are useful for feature engineering : $log(x)$, x^k , e^x



Other rescaling transformations

More scaling transformations are useful for feature engineering : ***log(x)*** , x^k , e^x

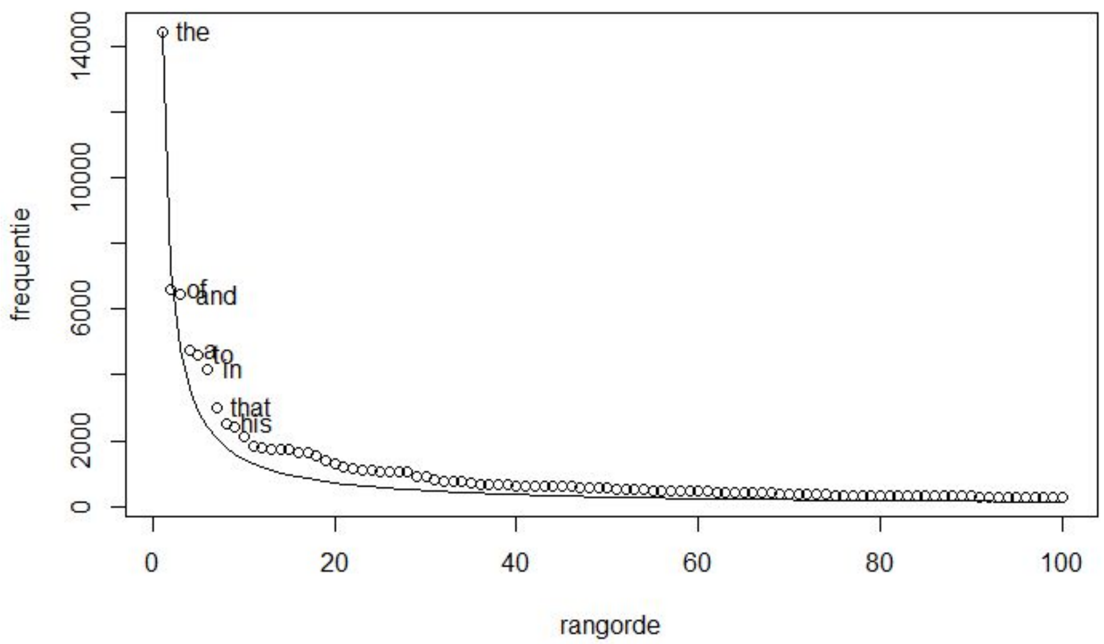


Logarithmic transformation

– **log-likelihood** (products→sums; easier to compute). Often used in LMs optimization (MLE)

– **Zipf's Law** (word frequency distributions)

1. the
2. and
3. of
4. a
5. to
6. that
7. this
8. …



Logarithmic transformation

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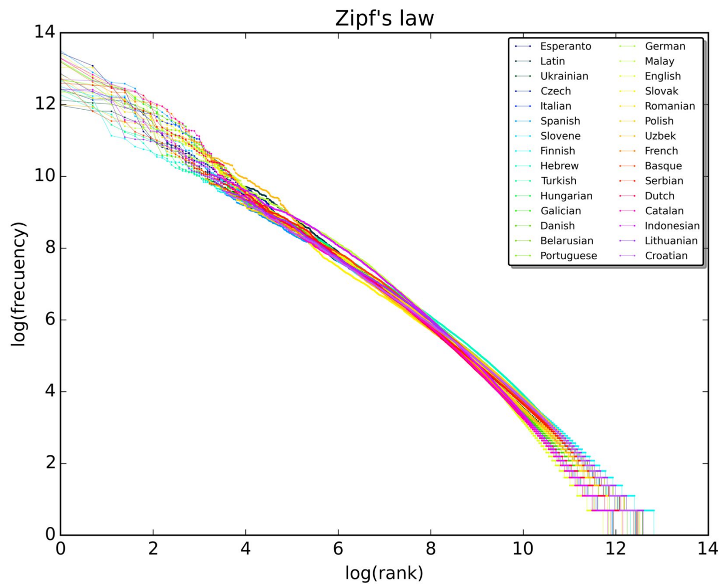


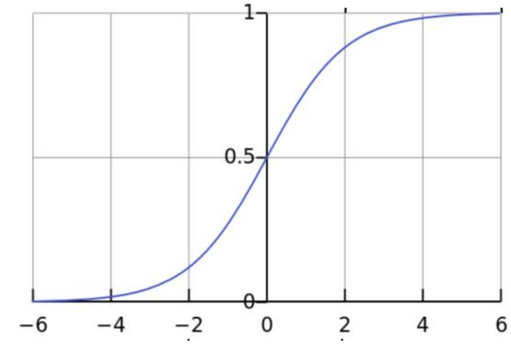
$$\text{word frequency} \propto \frac{1}{\text{word rank}}$$

Zipf's law plot for the first 10 million words in 30 Wikipedias

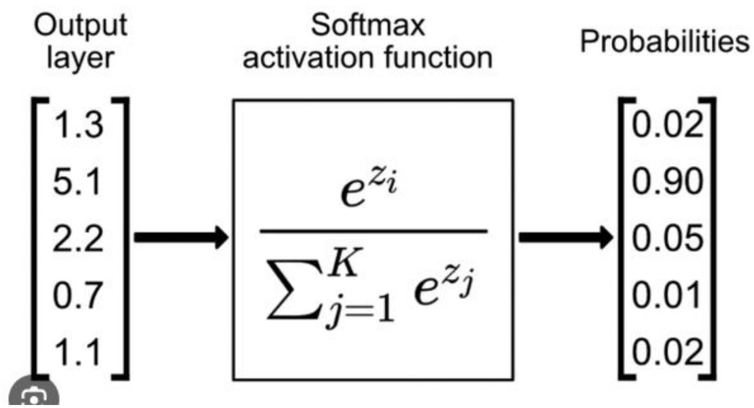
Sigmoid transformation

– **sigmoid**

$$A = \frac{1}{1+e^{-x}}$$



– **softmax**



Lecture outline

- Scaling
- **Imputation**
- Sampling

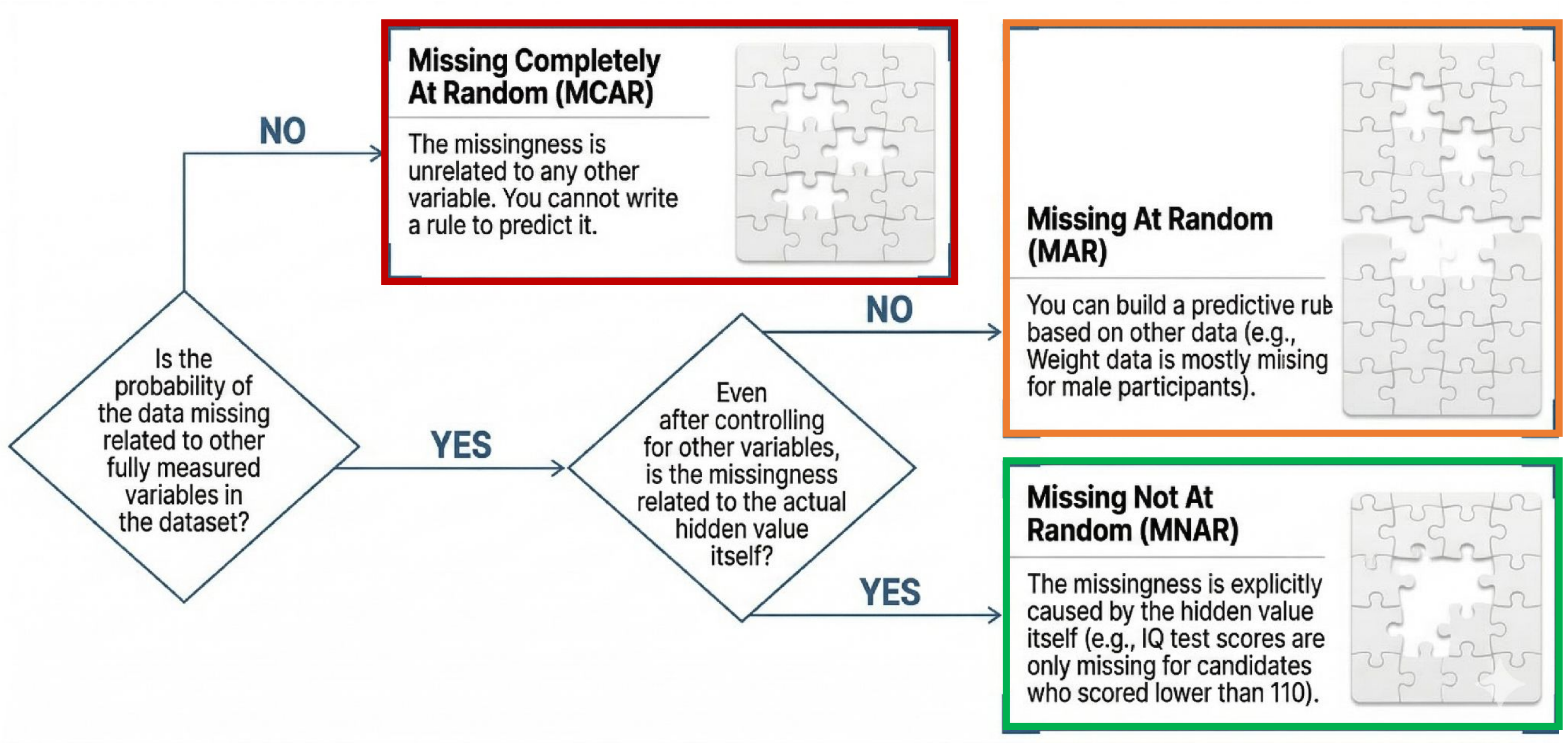
The Missing Data Dilemma

Datasets are rarely complete.

Before deciding how to fill a missing value, a data scientist must diagnose
the mechanism of its missingness.

What **causes** missing data: https://www.jstor.org/stable/2335739?seq=1 (Rubin, 1976)

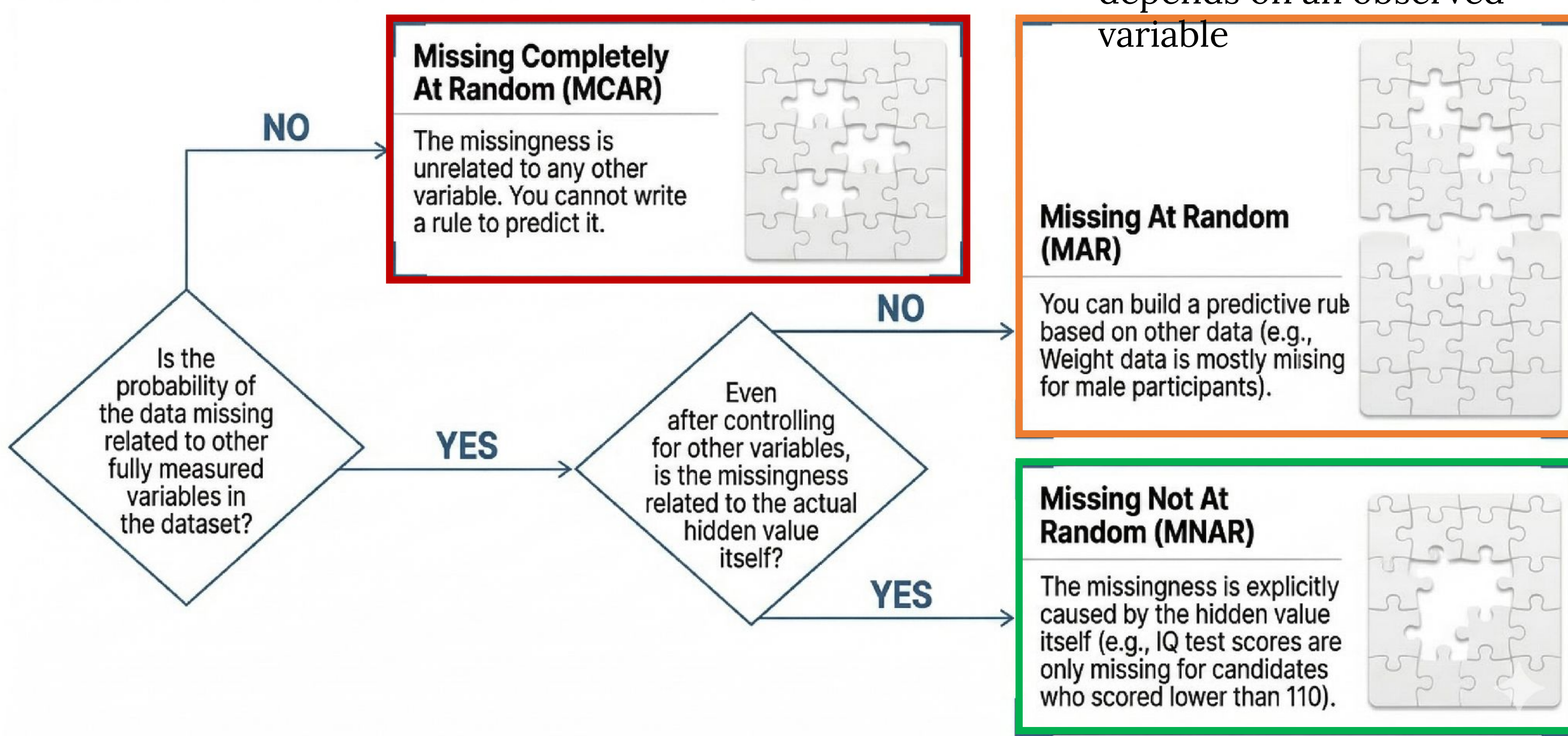
Diagnosing Missingness: The Decision Tree



Diagnosing Missingness: The Decision Tree

causes are unrelated, e.g. a broken device

depends on an observed variable



depends on the missing variable itself

Missing Completely At Random (MCAR)

Complete data	
Age	**IQ score**
25	133
26	121
29	91
30	105
30	110
31	98
44	118
46	93
48	141
51	104
51	116
54	97

Incomplete data	
Age	**IQ score**
25	
26	121
29	91
30	
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31	
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Missing data are **MCAR** when the probability of missing data on a variable is **unrelated** to any other measured variable and is **unrelated** to the variable with missing values itself.

You simply cannot make a rule to predict what's missing

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Missing data are **MCAR** when the probability of missing data on a variable is **unrelated** to any other measured variable and is **unrelated** to the variable with missing values itself.

You simply cannot make a rule to predict what's missing

- consider only complete data
- simple imputation (mean, median)

Missing At Random (MAR)

Gender	Weight
Male	81
Female	50
Male	
Male	
Female	66
Male	
Male	75
Male	
Female	

Missing data are **MAR** when the probability of missing data on a variable is **related** to other fully measured variables.

You can try to make a rule in the example:
"male weights are missing"
which may not be 100% accurate but is **better than a random guess.**

- predict under observed vars (regressions, MLE, ML models, k-NN, sample similarity)

Missing Not At Random (MNAR)

Complete data	
Age	**IQ score**
25	133
26	121
29	91
30	105
30	110
31	98
44	118
46	93
48	141
51	104
51	116
54	97

Incomplete data	
Age	**IQ score**
25	133
26	121
29	
30	
30	110
31	
44	118
46	
48	141
51	
51	116
54	

Data are **MNAR** when the missing values on a variable are **related** to the values of that variable itself, even after controlling for other variables.

That is, you can make rule that **accurately predicts** what's missing.

For example, when some IQ is missing, and only the people with low IQ values. Your rule is:
"missing If IQ is lower than 110"

- needs domain knowledge, assumptions, shared param models

Diagnosing Missingness

Q1: Students complete a survey about weekly study hours. First-year students are less likely to answer the question than later-year students. The dataset contains each student's year level.

Q2: Participants are asked to report their annual income. People with very high incomes are more likely to leave the question unanswered.

Q3: A weather station records temperature every hour. During a power outage, measurements are missing for six randomly affected hours. The outage was unrelated to the weather.

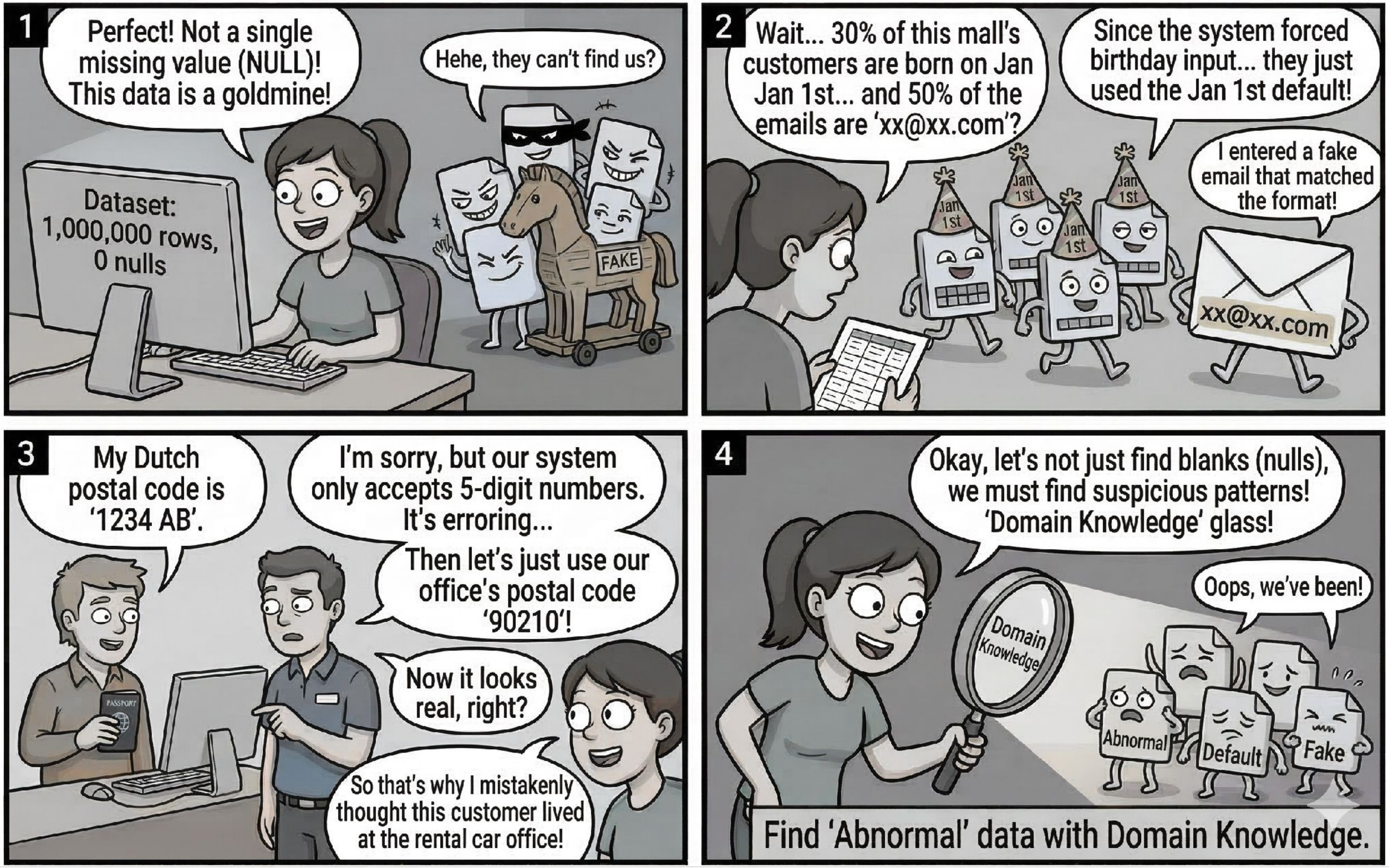
Diagnosing Missingness

Student	Tutorial group	Attendance	Exam score
A	Monday	90%	78
B	Tuesday	60%	Missing
C	Monday	85%	82
D	Tuesday	55%	Missing
E	Monday	70%	68

Disguised Missing Data

- Everyone's birthday is January 1st?
- Email address is xx@xx.com
- Adriaans and Zantige
  - "*Recently, a colleague rented a car in the USA. Since he was Dutch, his postal code did not fit the fields of the computer program. The car hire representative suggested that he use the code of the rental office instead.*"
- How to handle
  - Look for "unusual" or suspicious values in the dataset, using knowledge about the domain

Disguised Missing Data



Simple Imputation for numerical data

Imputation? methods to guess missing data

Statistical Measurement Imputation: simple strategies to fill out *numerical* data instances/records:

- Mean
- Median
- Mode

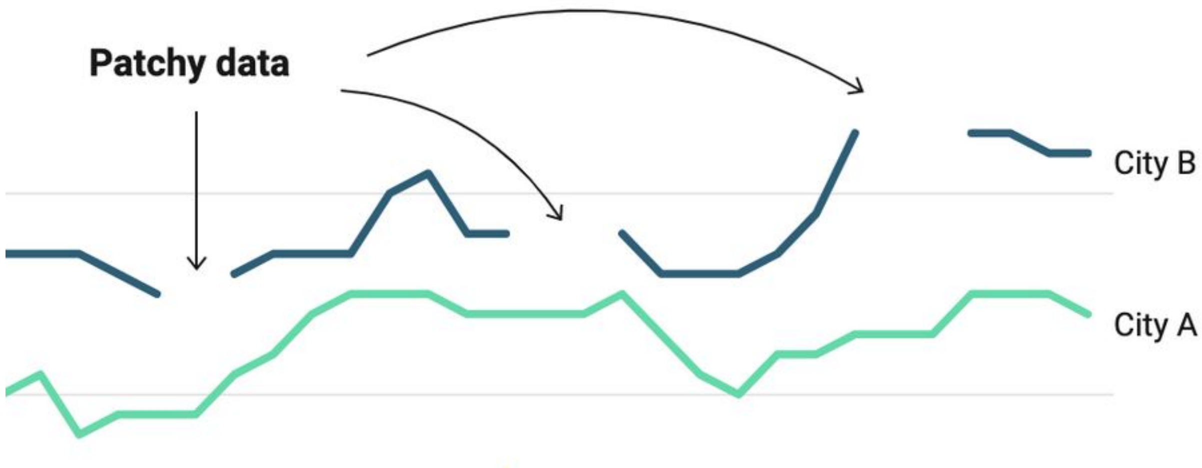
Pro: easy to compute and administer; no loss of records

Con: biases other statistical measurements

- variance
- standard deviation

What number to insert?

Different strategies suggest different values:



- Average of endpoints?
- Some sort of curved line?
- Most common/typical value?

Simple strategies with sklearn

Use numpy SimpleImputer to fill in the null values

```
>>> import pandas as pd
>>> from sklearn.impute import SimpleImputer
>>>
>>> import numpy as np
>>> formatter = "{:.2f}".format
>>> np.set_printoptions(formatter={'float_kind':formatter})
>>>
>>> text = '{"Caring for Pythons": [4.5, null, null, null, 4.0, 3.5, null, null, 4.5, 4.0],
 "Pythons 101: Must Know Things": [null, 4.5, 4.5, null, null, 4.0, 3.5, 3.0, null, 3.5],
 "Pythons: The Best Companions": [null, 5.0, null, null, null, 4.5, null, 3.5, 3.0, 4.0], "A
 Python Diet": [null, null, null, null, null, 3.0, 2.5, null, null, 3.5], "Python Cookbook":
 [null, null, null, null, null, 1.0, 0.0, null, 1.5, 0.5], "Slither": [4.5, null, 4.5, null,
 null, 5.0, 4.0,null, 4.5, null]}'
>>> data = pd.read_json(text)
>>>
>>> imp_mean = SimpleImputer(strategy='mean')
>>> imp_median = SimpleImputer(strategy='median')
>>> imp_mode = SimpleImputer(strategy='most_frequent')
>>>
```

Simple strategies with sklearn

mean
= 'average'

```
>>> imp_mean.fit(data)
SimpleImputer()
>>> imp_mean.transform(data)
array([[4.50, 3.83, 4.00, 3.00, 0.75, 4.50],
       [4.10, 4.50, 5.00, 3.00, 0.75, 4.50],
       [4.10, 4.50, 4.00, 3.00, 0.75, 4.50],
       [4.10, 3.83, 4.00, 3.00, 0.75, 4.50],
       [4.00, 3.83, 4.00, 3.00, 0.75, 4.50],
       [3.50, 4.00, 4.50, 3.00, 1.00, 5.00],
       [4.10, 3.50, 4.00, 2.50, 0.00, 4.00],
       [4.10, 3.00, 3.50, 3.00, 0.75, 4.50],
       [4.50, 3.83, 3.00, 3.00, 1.50, 4.50],
       [4.00, 3.50, 4.00, 3.50, 0.50, 4.50]])
```

median
= middle one

```
>>> imp_median.fit(data)
SimpleImputer(strategy='median')
>>> imp_median.transform(data)
array([[4.50, 3.75, 4.00, 3.00, 0.75, 4.50],
       [4.00, 4.50, 5.00, 3.00, 0.75, 4.50],
       [4.00, 4.50, 4.00, 3.00, 0.75, 4.50],
       [4.00, 3.75, 4.00, 3.00, 0.75, 4.50],
       [4.00, 3.75, 4.00, 3.00, 0.75, 4.50],
       [3.50, 4.00, 4.50, 3.00, 1.00, 5.00],
       [4.00, 3.50, 4.00, 2.50, 0.00, 4.00],
       [4.00, 3.00, 3.50, 3.00, 0.75, 4.50],
       [4.50, 3.75, 3.00, 3.00, 1.50, 4.50],
       [4.00, 3.50, 4.00, 3.50, 0.50, 4.50]])
```

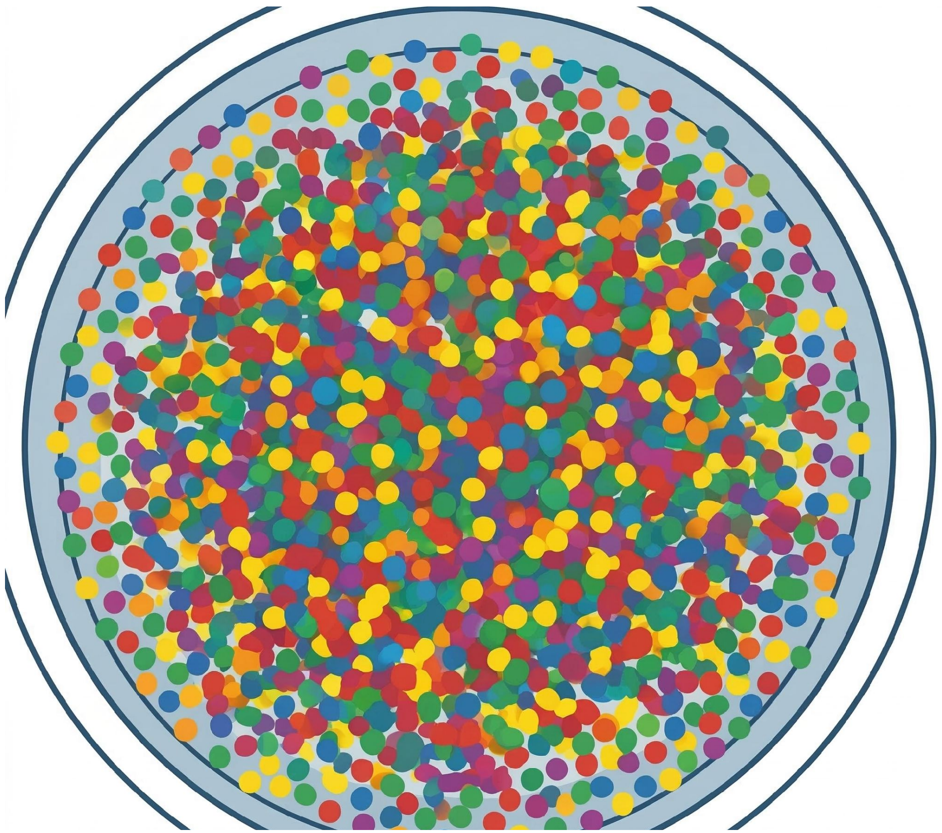
most frequent

```
>>> imp_mode.fit(data)
SimpleImputer(strategy='most_frequent')
>>> imp_mode.transform(data)
array([[4.50, 3.50, 3.00, 2.50, 0.00, 4.50],
       [4.00, 4.50, 5.00, 2.50, 0.00, 4.50],
       [4.00, 4.50, 3.00, 2.50, 0.00, 4.50],
       [4.00, 3.50, 3.00, 2.50, 0.00, 4.50],
       [4.00, 3.50, 3.00, 2.50, 0.00, 4.50],
       [3.50, 4.00, 4.50, 3.00, 1.00, 5.00],
       [4.00, 3.50, 3.00, 2.50, 0.00, 4.00],
       [4.00, 3.00, 3.50, 2.50, 0.00, 4.50],
       [4.50, 3.50, 3.00, 2.50, 1.50, 4.50],
       [4.00, 3.50, 4.00, 3.50, 0.50, 4.50]])
```

Lecture outline

- Scaling
- Imputation
- **Sampling**

What if there's too much data?



Sampling from data

Sampling = "Selecting **a few** examples to represent **all** the data"

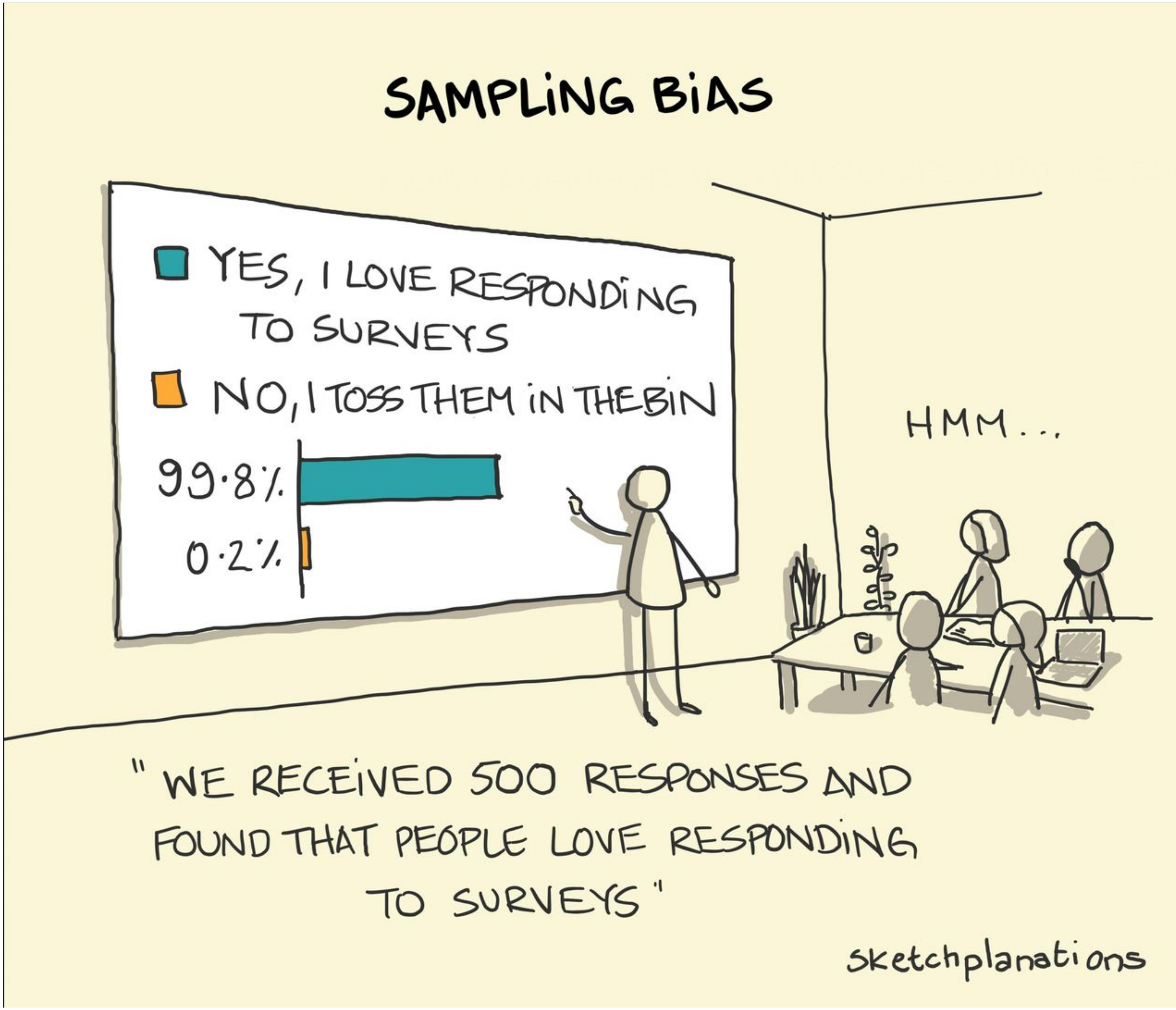
When to sample?

When the dataset is too large to efficiently analyse in full.
Identifying and analysing a representative sample is more efficient and cost-effective than surveying all the data.

How much data to select?

Too little means you will have *sampling error* and the results will not be trustworthy

- Different data and different needs require different sizes


SAMPLING BIAS
YES, I LOVE RESPONDING TO SURVEYS
NO, I TOSS THEM IN THE BIN
99.8%
0.2%
HMM...
" WE RECEIVED 500 RESPONSES AND FOUND THAT PEOPLE LOVE RESPONDING TO SURVEYS "
sketchplanations

On the importance of sampling

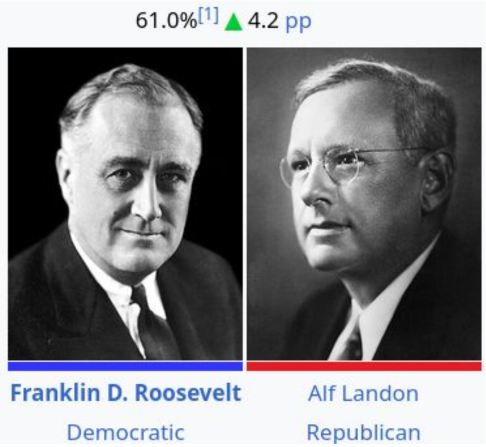
https://en.wikipedia.org/wiki/1936_United_States_presidential_election

	Literary Digest poll	Actual election result
Alf Landon	57% — predicted winner	36.5%
Franklin D. Roosevelt	43%	60.8% — actual winner

Literary Digest: ~10 million questionnaires and received around 2.4 million responses. However, the names were largely obtained from:

- telephone directories;
- automobile-registration records;
- the magazine's subscriber list

Wealthy people



There was also **non-response bias:** only about 24% of those contacted replied, and the people who chose to reply were politically different from those who did not.

Sampling methods

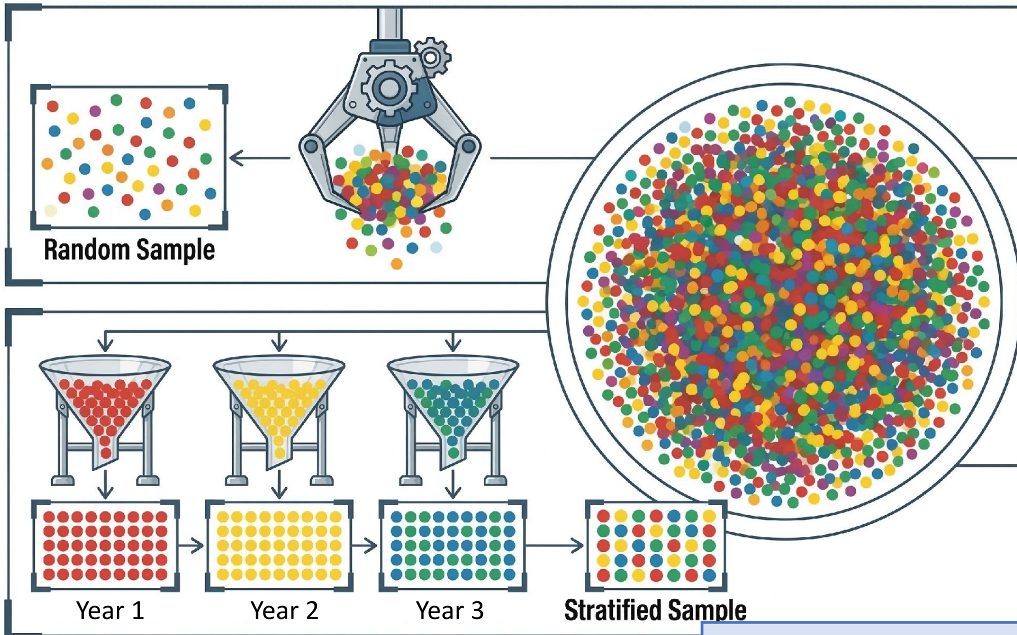
Who will win the next election?
You cannot ask everyone!

- **Random sampling**: You randomly pick 10,000 phone numbers from the phone book and call
  - Without/with replacement: what you do if no-one answers
- **Stratified sampling**: You split the population into groups depending on the relevant features:
  - age groups / gender / income levels
- Then you call within each group. Can choose randomly, or make another split first...

Example: Survey on University Education

The collected responses should represent the true distribution of opinions across all year levels.

If the survey is conducted in places where students stay the longest, such as near cafeterias or common areas, first-year students may dominate the sample.



Approach 1: Random Sampling

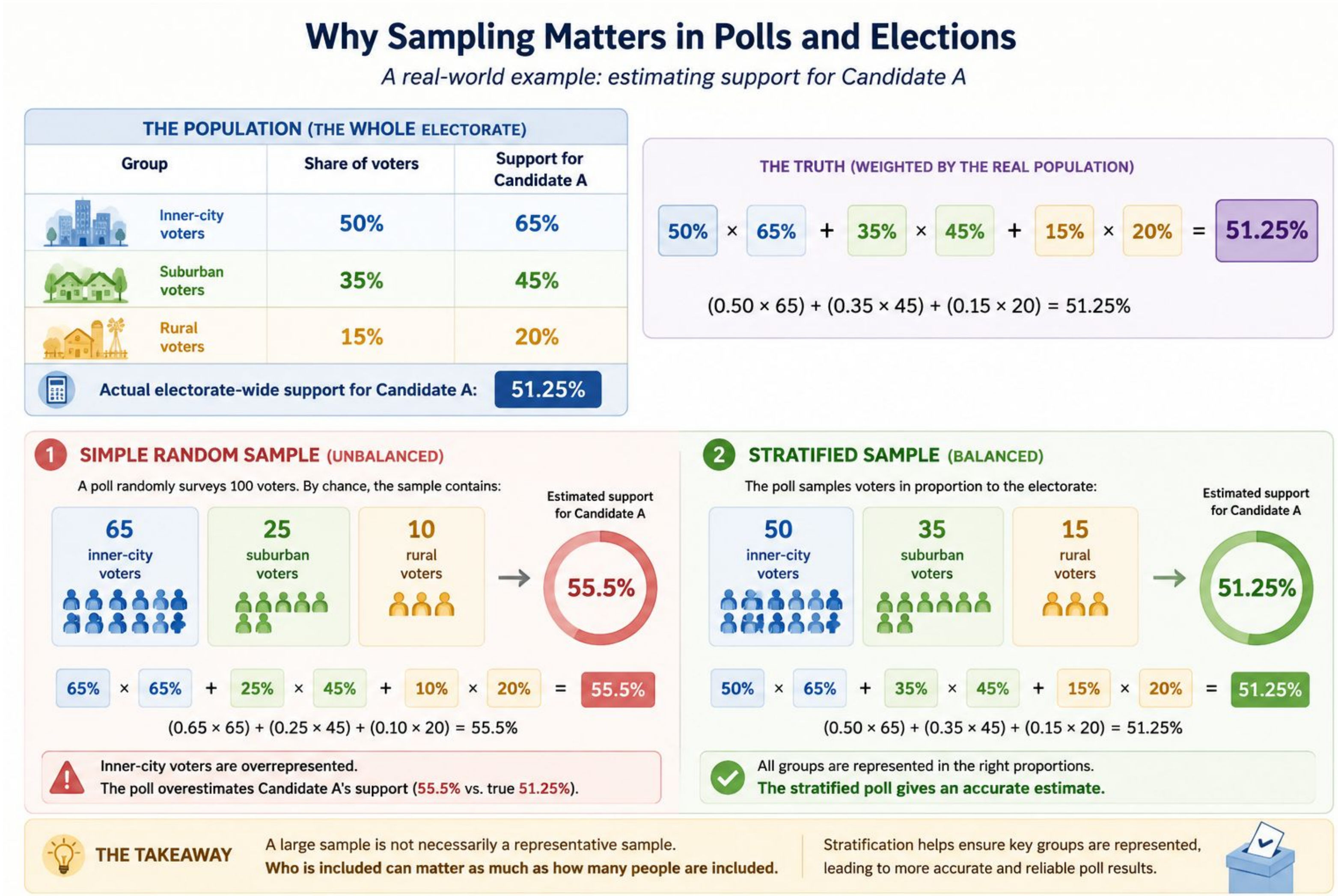
Distributing surveys randomly to students (e.g., handing out questionnaires in front of the campus cafeteria). Might miss small/rare groups

Approach 2: Stratified Sampling

Divide the student population into clear groups based on year level. Then sample students from each group in proportion to their actual share of the total student population.

Different year levels experience the university differently. Stratified sampling ensures that even smaller groups (such as Year 4 students) are properly represented and their perspectives

Example: Elections



Sampling challenges

Challenge 1:

Sample must be

- **Representative**: Sample has the same important properties as the full population
- **Balanced**: Sample does not exclude some cases

Challenge 2:

Sample must be **large enough** to be (reasonably) trustworthy

- Heavy statistics needed to ensure this!
- And even then, the subjects may not report honestly (this is common with politics)

Week 3 Workshop & Lecture 2 Alignment

Week 3 Workshop (Hands-on Practice)

- **Handling Missing Data:** Identifying missing values and applying simple imputation methods.
- **Data Quality Validation:** Detecting semantic, range, and format errors in datasets.
- **Sampling for Imbalance:** Using undersampling or oversampling to balance class distributions.

Key Learning Outcome

- Connect **preprocessing concepts from Lecture 2** with **practical data cleaning and sampling techniques** implemented in the workshop.